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Bernstein Vazirani

Video: Introduction to Bernstein Vazirani1 min

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Graded Assignment: Assignment 1Submitted

Your grade: 100%

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Assessment 1

Next item →

1. Which quantum gate is used to create a superposition of qubit states?1 / 1 point

☒ Hadamard gate

☐ Toffoli gate

☐ Pauli-X gate

☐ CNOT gate

COACH

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice

Let's chat

Correct

Correct! The Hadamard gate is used to create a superposition of qubit states.

Assignment details

Due

Dec 16, 11:59 PM IST

Attempts

Unlimited

2. Which gate would you use in a circuit to create entanglement between two qubits?1 / 1 point

☐ Hadamard and Toffoli gates

☒ Hadamard and CNOT gates

☐ Pauli-X and Hadamard gates

☐ CNOT and Toffoli gates

COACH

To pass you need at least 80%. We keep your highest score.

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Correct

Correct! The combination of Hadamard and CNOT gates is used to create entanglement between two qubits.

Assignment details

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3. What is the primary function of the CNOT gate in quantum circuits?1 / 1 point

☐ To create superposition of qubit states

☐ To perform a controlled-controlled-NOT operation

☐ To flip the state of a single qubit

☒ To create entanglement between qubits

Correct

Correct! The CNOT gate is primarily used to create entanglement between qubits.

4. What is the significance of the ZGate in quantum computing?1 / 1 point

☐ It measures the state of a qubit

☒ It flips the phase of the qubit state

☐ It entangles two qubits

☐ It flips the state of a qubit from $|0\rangle$ to $|1\rangle$

Correct

Correct! The ZGate flips the phase of the qubit state, turning $|+\rangle$ into $|-\rangle$.

5. Which of the following operations are part of the Bernstein-Vazirani algorithm?1 / 1 point

☒ Applying Hadamard gates to each qubit

Correct

Correct! The Bernstein-Vazirani algorithm starts by applying Hadamard gates to each qubit to create superposition.

☒ Measuring qubits after applying gates

Correct

Correct! Measurement is a crucial step after applying the necessary gates in the Bernstein-Vazirani algorithm.

☒ Applying a series of CNOT gates

Correct

Correct! CNOT gates are used in the Bernstein-Vazirani algorithm to entangle qubits.

☐ Applying Grover's algorithm

☐ Using quantum tunneling